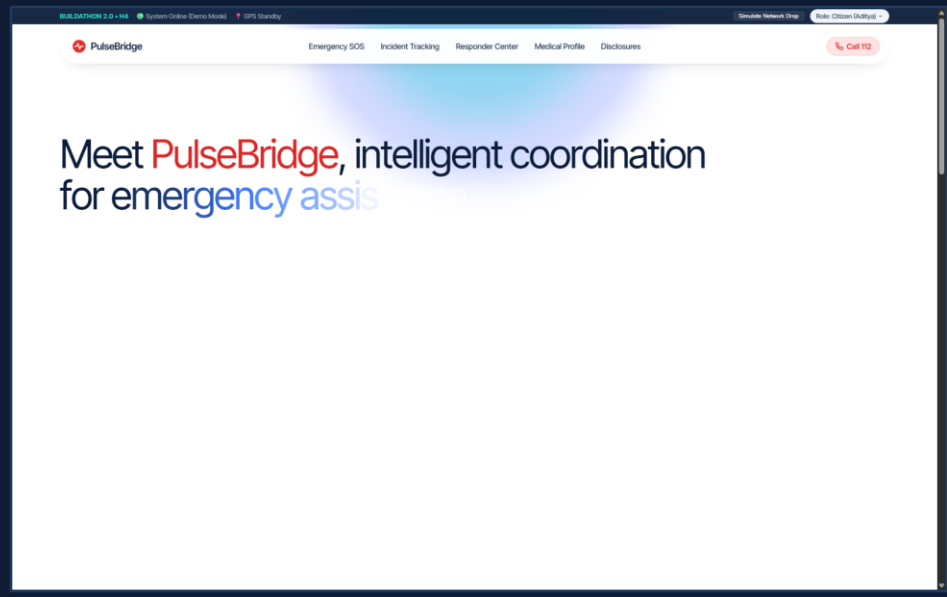
Emergency Response & Coordination Platform

Connecting emergency reporting, incident tracking, and tactical responder coordination.

PROJECT & TEAM DETAILS (EDITABLE PLACEHOLDERS)

* Team Name: [Your Team Name / Team ID]
* Presenters: [Lead Presenter, Member 2, Member 3]
* Institution: [College / University / Organization Name]
* Hackathon Track: H4 - Emergency Health Assistance
* Live Web Platform: shrihari12012007-web.github.io/Pulse-Bridge/
* GitHub Repository: github.com/shrihari12012007-web/Pulse-Bridge
* Prototype Status: Fully Functional Interactive Web & CAD Demo



INTEGRATED EMERGENCY RESPONSE SUITE

1-Tap SOS Dispatch
False-alarm buffer + 112 link

GPS & Indoor Landmark
Sensory fix + room/floor notes

Realtime Ref ID Tracking
Zero-login transparent updates

Tactical CAD Dashboard
Leaflet map & unit dispatch

PROBLEM STATEMENT & ANALYSIS

Critical bottlenecks in campus, industrial, and community crisis response

01. Delayed Incident Reporting

CRITICAL 5-10 MIN LOSS

* Panic and shock prevent callers from finding phone numbers or explaining symptoms.
* Bystanders hesitate or assume someone else has called, losing the vital Golden Hour window.
* Phone lines are frequently busy or require lengthy manual address verification.

02. Incomplete Location Information

INDOOR LOCATION DRIFT

* GPS signals fail or drift by 50+ meters inside multi-story concrete buildings and labs.
* Responders receive vague locations ("somewhere in Block B") with no floor or room number.
* Emergency teams waste 4 to 8 minutes searching corridors instead of delivering care.

03. Difficulty Coordinating Responders

NO UNIFIED CAD CONSOLE

* Response teams rely on uncoordinated phone calls, chat groups, or verbal shouts.
* No centralized Computer-Aided Dispatch (CAD) console to triage priority or assign units.
* Leads to duplicate dispatches to one incident while another critical patient is unattended.

04. Lack of Clear Status Visibility

ANXIETY & BLIND SPOTS

* Citizens receive zero confirmation on whether their emergency alert was acknowledged.
* Lack of feedback triggers extreme anxiety and repeated duplicate emergency calls.
* No audit trail exists to evaluate response latency or responder accountability.

HACKATHON INSIGHT: Over 80% of emergency response delay occurs in the first 5 minutes due to manual reporting, lost indoor context, and uncoordinated dispatch. PulseBridge directly eliminates these failure points.

THE PULSEBRIDGE PLATFORM

Bridging emergency reporting, location telemetry, tracking, and tactical responders

CITIZEN REPORTING

Rapid SOS & Hotlines

* Single-Tap SOS Trigger: Large accessible button with 3s cancel buffer to prevent false alarms.

* Direct 112 Hotline: Permanent direct dial button to official emergency services, never blocked by login.

* Consented Health Snapshot: Securely attaches blood group, allergies, and emergency contacts.

LOCATION & TRACKING

Dual Location Engine

* GPS Sensor Telemetry: Captures precise latitude, longitude, and accuracy radius confidence.

* Indoor Landmark Fallback: Room, floor, wing, and building notes for GPS-denied environments.

* Unique Reference Token: Issues trackable reference IDs (e.g. PB-2026-9812) for zero-barrier updates.

RESPONDER DISPATCH

Tactical CAD Console

* Computer-Aided Dispatch: Dark Navy tactical dashboard with prioritized triage queues (Critical to Low).

* Interactive Leaflet Map: Color-coded status pins showing caller locations, accuracy, and landmark notes.

* CAD State Machine: Strict status progression: Submitted -> Acknowledged -> Assigned -> Resolved.

SIMPLE CLOSED-LOOP WORKFLOW

Citizen in Crisis
SOS Trigger
->
Incident Record
Ref ID + Telemetry
->
CAD Triage
Acknowledge
->
Responder Unit
Unit Assigned
->
Resolution
Audit Logged

USER JOURNEY

Seamless operational path from emergency trigger to on-site medical resolution

STEP 01 | TIME: < 5 SEC

Report Emergency

* Citizen triggers 1-tap emergency SOS on mobile or desktop browser.

* 3-second audible cancellation buffer prevents accidental activation.

* Direct 112 hotline fallback link is permanently available.

STEP 02 | LOCATION FIX

Record Details

* Browser Geolocation sensor captures latitude, longitude, and accuracy radius.

* Indoor landmark fallback allows instant entry of building, floor, or room.

* User consented medical profile (blood group, allergies) securely attached.

STEP 03 | CAD DISPATCH

Review & Dispatch

* Incident alerts Tactical Responder Console with immediate audio-visual cue.

* Dispatcher reviews caller severity, medical snapshot, and map coordinates.

* Dispatcher acknowledges ticket and assigns response team / paramedic unit.

STEP 04 | AUDIT LOGGED

Track & Resolve

* State machine updates: SUBMITTED -> ACKNOWLEDGED -> ASSIGNED -> RESOLVED.

* Citizen monitors live progress transparently via Reference ID (PB-2026-9812).

* Immutable audit log records all timestamps and responder actions to resolution.

INCIDENT LIFECYCLE STATE MACHINE

[SUBMITTED] ---> [ACKNOWLEDGED by Dispatcher] ---> [ASSIGNED to Response Unit] ---> [RESOLVED with Audit Log]

PLATFORM CAPABILITIES

Six core capabilities designed for life-saving speed, accuracy, and operational clarity

EMERGENCY SOS

Emergency SOS & 112 Fallback

* Instant 1-tap activation modal with 3s cancel window.
* Direct dial-112 hotline never blocked by logins or offline errors.
* Immediate visual alert triggers across CAD network.

TELEMETRY

GPS & Manual Coordinates

* HTML5 Geolocation sensor captures latitude, longitude, and accuracy.
* Manual coordinate override and indoor floor/room landmark fallback.
* Purpose-built for multi-story campus buildings where GPS drifts.

TRANSPARENCY

Incident Tracking (Ref ID)

* Generates unique reference token (e.g. PB-2026-9812).
* Zero account creation required for citizens to track rescue status.
* Real-time visual badges show triage updates and responder assignments.

CAD CONSOLE

Tactical Responder Dashboard

* Dark Navy CAD dashboard optimized for emergency duty dispatchers.
* Interactive Leaflet OpenStreetMap with colored priority marker pins.
* Instant incident search, status filtering, and live metric cards.

WORKFLOW

Acknowledge, Assign & Resolve

* Strict role-based status transitions eliminate coordination confusion.
* One-click action to acknowledge incident and assign field responders.
* Immutable, timestamped audit log preserved for accountability.

HEALTH DATA

Consent-Based Medical Profile

* User-controlled blood group, severe allergies, and chronic medications.
* Encrypted and shared strictly upon active SOS dispatch.
* Empowers first responders to administer safe, informed triage on arrival.

TECHNICAL ARCHITECTURE

Inspected repository source code: genuine technologies, zero invented integrations

REACT 19 + TS

Frontend Framework & Routing

React 19 | TypeScript | Vite | React Router v7 (HashRouter for zero-config GitHub Pages hosting)

LEAFLET + OSM

Interactive Maps & Geolocation

Leaflet 1.9 & OpenStreetMap (react-leaflet) | Custom SVG status pins | HTML5 Geolocation API

POSTGRES + RLS

Backend, Database & Storage

Supabase Client | PostgreSQL 15+ | Row-Level Security (RLS) policies | Realtime Replication | Node.js Server

CI/CD & OFFLINE

Resilience & Deployment

Service abstraction layer | Truthful offline detection | Local draft store | GitHub Actions CI/CD | Render

SYSTEM DATA FLOW ARCHITECTURE

CITIZEN MOBILE & WEB CLIENT (LIGHT THEME)
*** 1-Tap SOS Activation | 3s Cancel Buffer | India 112 Permanent Hotline**
*** Browser Geolocation Sensor Telemetry | Indoor Room/Floor Landmark Fallback**

DATA SERVICE ABSTRACTION LAYER (RESILIENCE)
* Unified Incident Service: Transparently switches between Supabase & Demo Store
* Truthful Offline Outage Detector: Alerts user to dial 112 when disconnected
* Local Draft Preservation & State Machine Validation

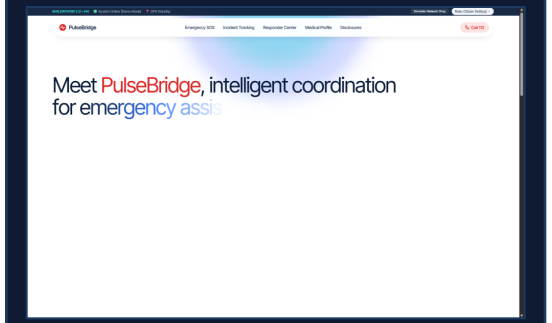
SUPABASE CLOUD / LOCAL DEMO STORE (POSTGRESQL 15+)
* Incidents Table: Priority, Coordinates, Landmark, Consented Health Snapshot
* Immutable Audit Events Log | Row-Level Security (RLS) Profile Protection
* Postgres Realtime Replication Channels

RESPONDER CAD DASHBOARD (DARK NAVY THEME)
*** Live Triage Queue: Critical, High, Medium, Low Filtering**
*** Interactive Leaflet Map with Real-time GPS Pins & Indoor Landmark Details**
*** Strict CAD State Transitions: Acknowledge -> Assign Unit -> Resolve**

Verified 100% against repository source code. No unconfigured integrations.

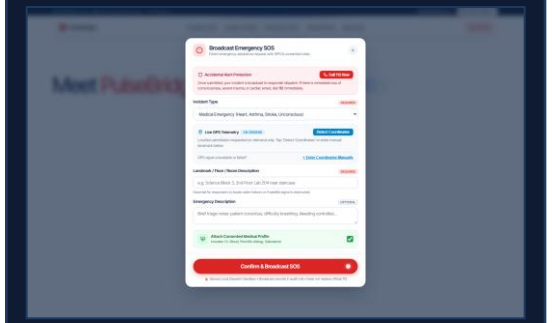
PROTOTYPE VALIDATION

Genuine application screenshots captured directly from active PulseBridge deployment



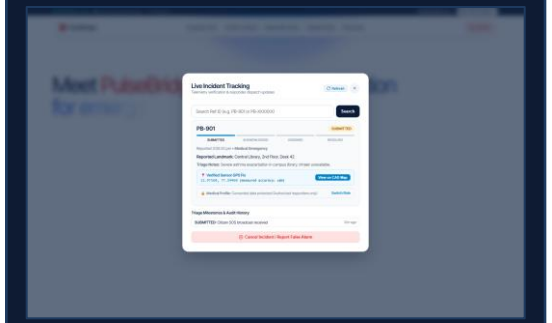
CITIZEN LANDING

1. Homepage
1-Tap SOS & 112 Hotline



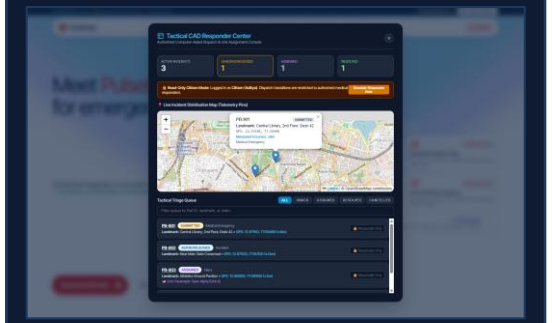
INCIDENT DISPATCH

2. Emergency SOS Form
GPS Telemetry & Landmark



LIVE TRACKING

3. Incident Tracking
Ref ID & Audit Timeline



CAD DASHBOARD

4. Responder Center
Leaflet Map & Triage Queue

IMPLEMENTED DEMO WORKFLOW (VERIFIABLE DURING LIVE DEMO)

1. SOS Submission -> 2. Reference ID Generated -> 3. Responder Queue Alert -> 4. Dispatcher Assignment -> 5. Case Resolution

MANDATORY PROTOTYPE DISCLOSURE & DEMO STATEMENT

PulseBridge is an engineering prototype built for BUILDATHON 2.0. The current workflow utilizes local demo storage and an event bus to demonstrate end-to-end responder coordination. It does NOT connect to municipal emergency dispatch or live ambulance fleets. In real life emergencies across India, dial 112 immediately.

ROADMAP & CONCLUSION

Strategic pathway from hackathon prototype to enterprise emergency coordination network

STRATEGIC FUTURE IMPROVEMENTS

1. **Reliable GPS & Location Verification**
   Integrate Wi-Fi RTT, Bluetooth Low Energy (BLE) beacons, and building floorplans for sub-meter indoor positioning.

2. **Live Backend Synchronization**
   Deploy production WebSocket clusters and conflict-free replicated data types (CRDTs) for seamless offline-to-online sync.

3. **Secure Responder Authentication**
   Institutional credential verification, paramedic licensing checks, and biometric authentication for dispatchers.

4. **Verified Emergency-Service (112) Integrations**
   Establish official Computer-Aided Dispatch (CAD) API bridges with municipal public safety answering points.

5. **Real-Time Notifications & Communication**
   Automated SMS/WhatsApp family crisis alerts, WebPush notifications, and low-latency WebRTC responder voice intercom.

PulseBridge

"PulseBridge - Connecting Every Emergency to a Coordinated Response."

Thank You Judging Panel & Mentors!
We welcome your questions and live demonstration.

BUILDATHON 2.0 | KNEW-2K26
Problem Statement: H4 - Emergency Health Assistance

Team Details (Editable):
* Presenters: [Lead Presenter, Member 2, Member 3]
* Institution: [College / University / Organization Name]

* Web Demo: shrihari12012007-web.github.io/Pulse-Bridge/
* GitHub: github.com/shrihari12012007-web/Pulse-Bridge